

Kernel-Level Access Control

Advanced Security in Linux Architectures for Cloud-Native
Environments

Aluthge M.N | IT24102268

2nd Year, 2nd Semester | IE2092: introduction to Cyber Security

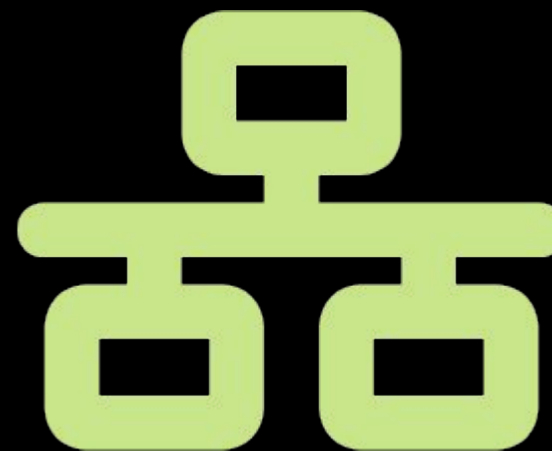
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The Security Challenge

The Perimeter is Gone.

Moving from physical servers to ephemeral microservices fundamentally breaks legacy security models.

- Containers share the same underlying kernel.
- A single breach exposes the entire host.
- Defense **must** shift to the OS core.



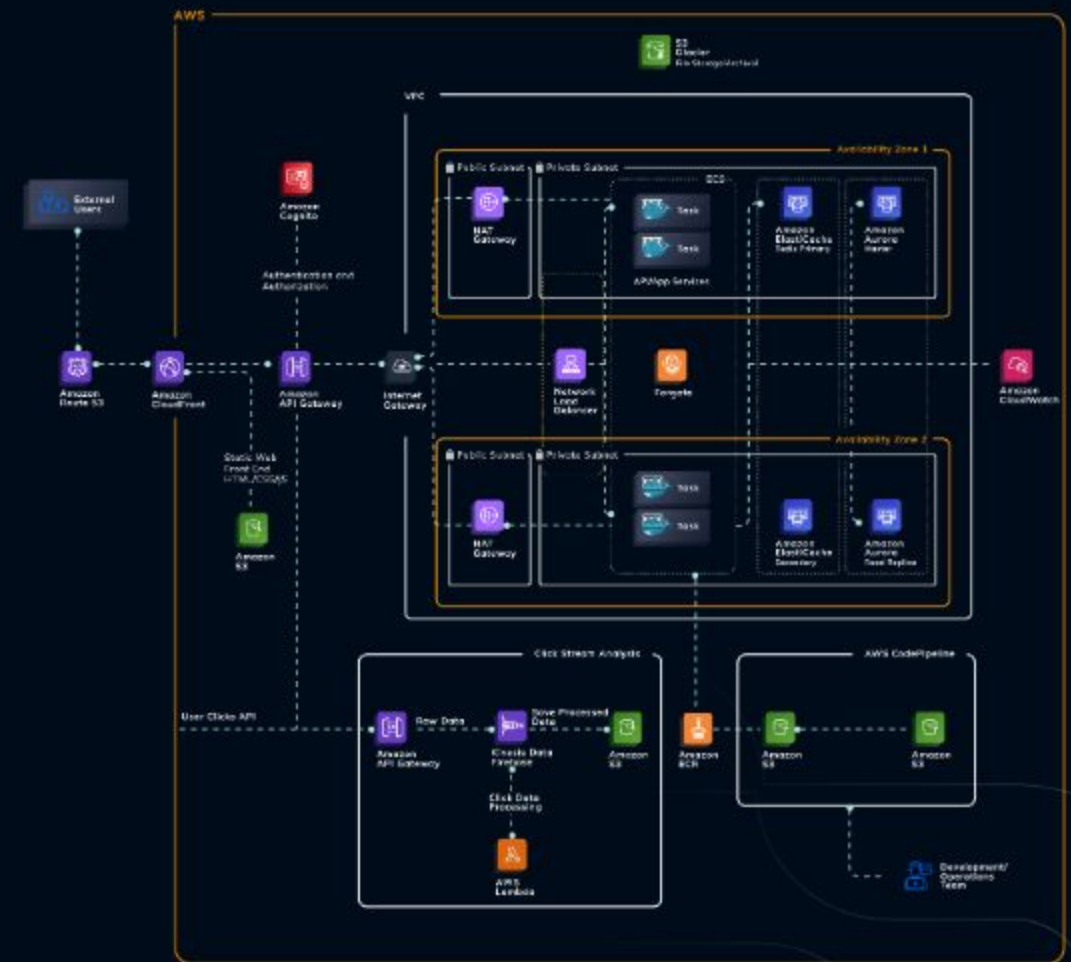
The Cloud-Native Paradigm Shift

Modern organizations have forsaken monolithic architectures in favor of microservices, splitting huge applications into thousands of ephemeral containers.

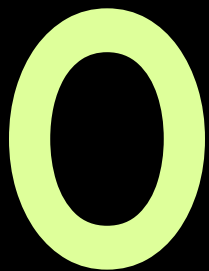
- Rapid spin-up/terminate cycles via Kubernetes.
- Shared OS kernel increases the attack surface.
- Legacy edge firewalls are now largely irrelevant.



Cloud Native Architecture Diagram



The Absolute Power of Ring 0



KERNEL PROTECTION RING

Unrestricted Access

The kernel acts as the absolute dictator of resources. If an attacker gains execution in **Ring 0**, all boundaries (namespaces, permissions, container walls) become meaningless.

Evolution of Access Paradigms



DAC

Discretionary models trust the user. Vulnerable to "Confused Deputy" attacks and permission inheritance flaws.



MAC

Mandatory models enforce global policy. Subjects and objects are labeled, removing power from individual users.



Zero Trust

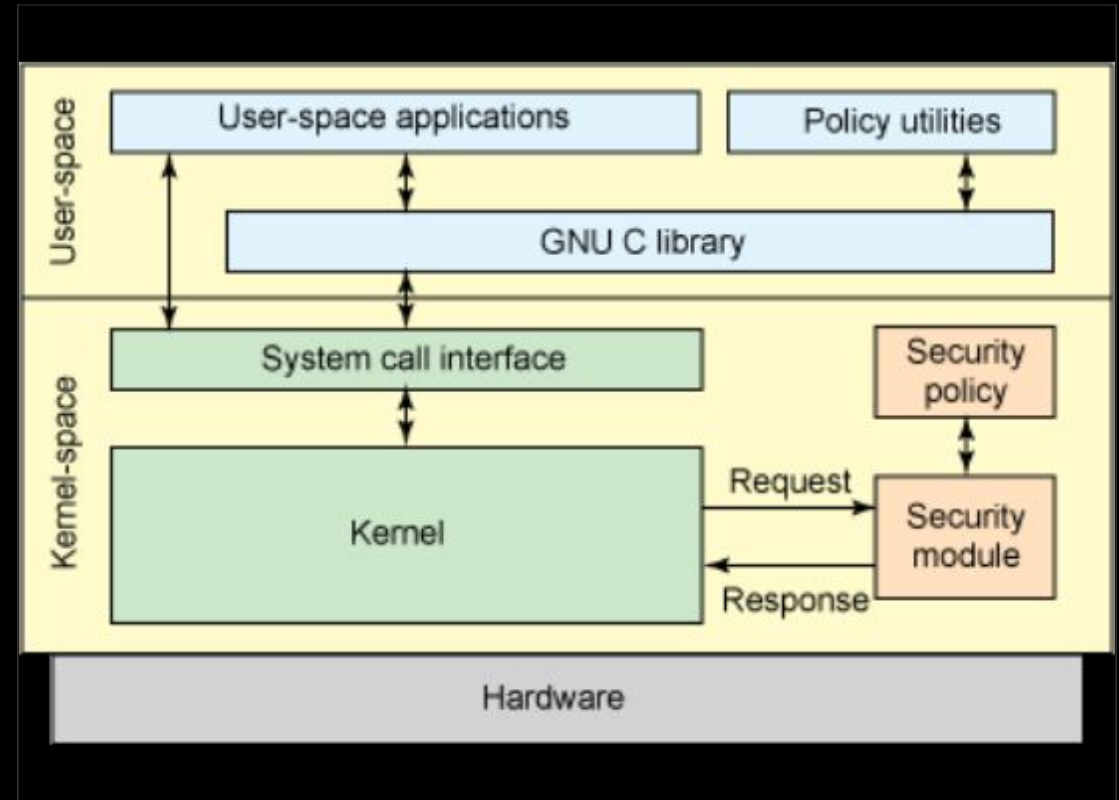
The system no longer trusts based on identity. Every system call is verified against a master in-kernel policy database.

LSM & The TOCTOU Fix

Strategic Inline Hooking

The **Linux Security Module (LSM)** framework injects generic hooks deep into execution paths.

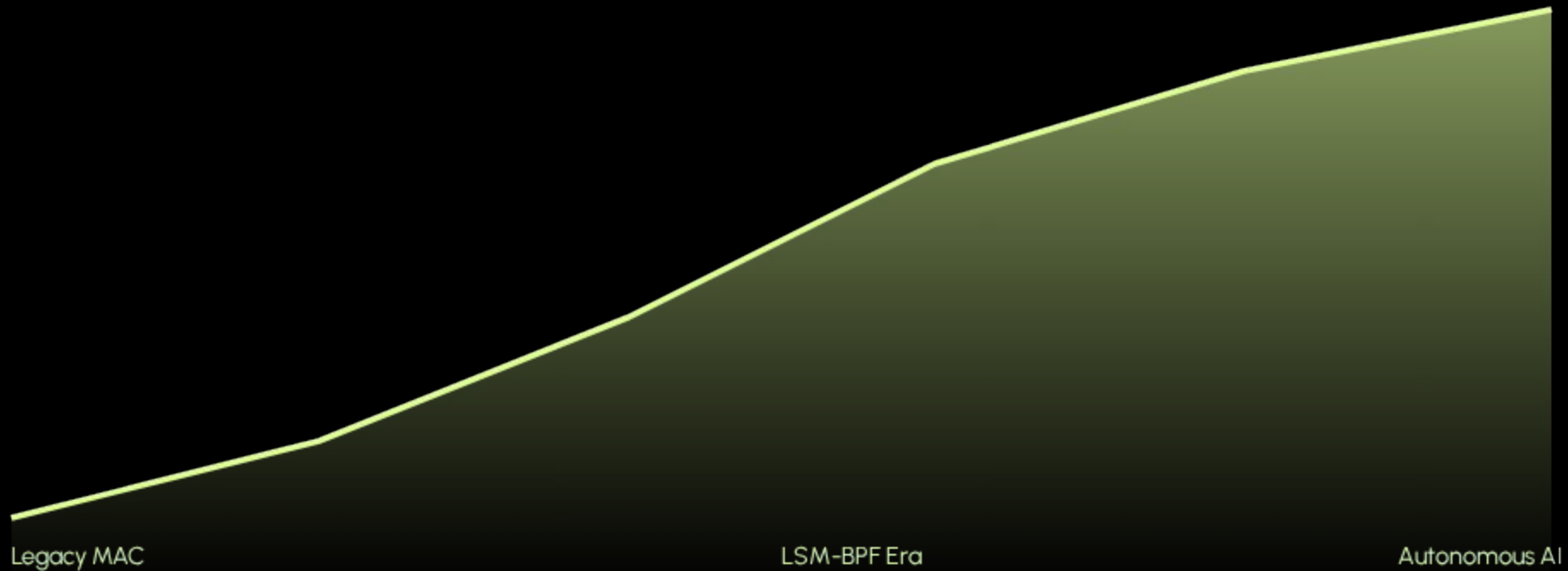
- Abolishes Time-of-Check to Time-of-Use (TOCTOU).
- Verification happens on tamper-proof references.
- Allows pluggable modules like SELinux.



SELinux vs. AppArmor

Feature	SELinux (NSA)	AppArmor (Ubuntu)
Primary Model	Label-Based (Type Enforcement)	Path-Based Confinement
Complexity	High (Steep learning curve)	Low (Human-readable profiles)
Performance Penalty	6% - 15% (Higher overhead)	< 2% (Lightweight)
Strongest Feature	Multi-Level Security (MLS)	Fast Learning/Complain Mode

The Rise of eBPF Programmability



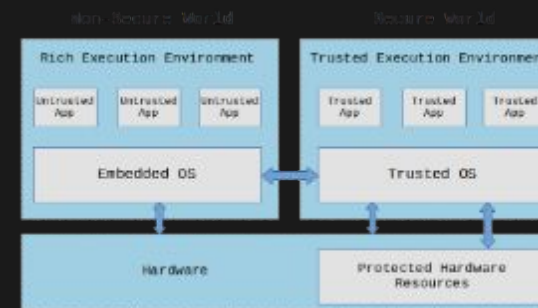
eBPF moves beyond static rules, introducing a sandboxed VM in the kernel for real-time enforcement and observability.

Defeating Code-Reuse Attacks



PAC

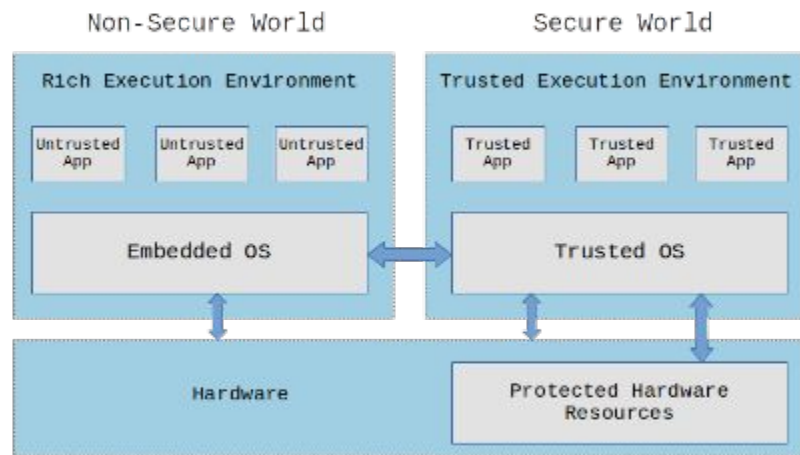
Pointer Authentication Codes use secret keys to cryptographically sign pointers, preventing ROP attacks.



BTI

Branch Target Identification enforces "landing pads" in hardware to stop malicious jumps (JOP).

HIVE: Hardware-Assisted Isolation

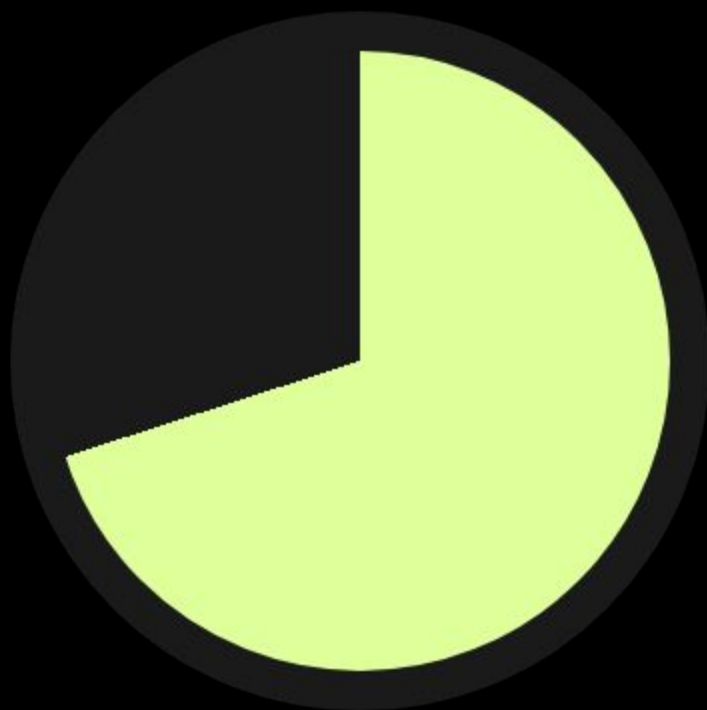


Immune Guardians

HIVE decouples the eBPF runtime from the host OS physically using **Trusted Execution Environments (TEEs)**.

- Actual security logic runs in a hardware-encrypted domain.
- Even a rootkit-compromised OS cannot alter bytecode.
- The hardware refuses "turn-off" instructions from a malicious OS.

The In-Kernel AI Revolution



■ Automated AI Response

■ Human Manual Tuning

AI algorithms now run directly on system call telemetry, learning behavioral baselines and blocking zero-day anomalies in nanoseconds.

Thank You!

Kernel-Level Access Control & Advanced Linux Security

Aluthge M.N (IT24102268)

Sri Lanka Institute of Information Technology

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Image Sources



<https://www.clickittech.com/wp-content/uploads/2024/07/Cloud-Native-Architecture-Diagram.png>

Source: www.clickittech.com



https://miro.medium.com/O*EsaeBLBqJqbYuRUP

Source: [medium.com](https://miro.medium.com)



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